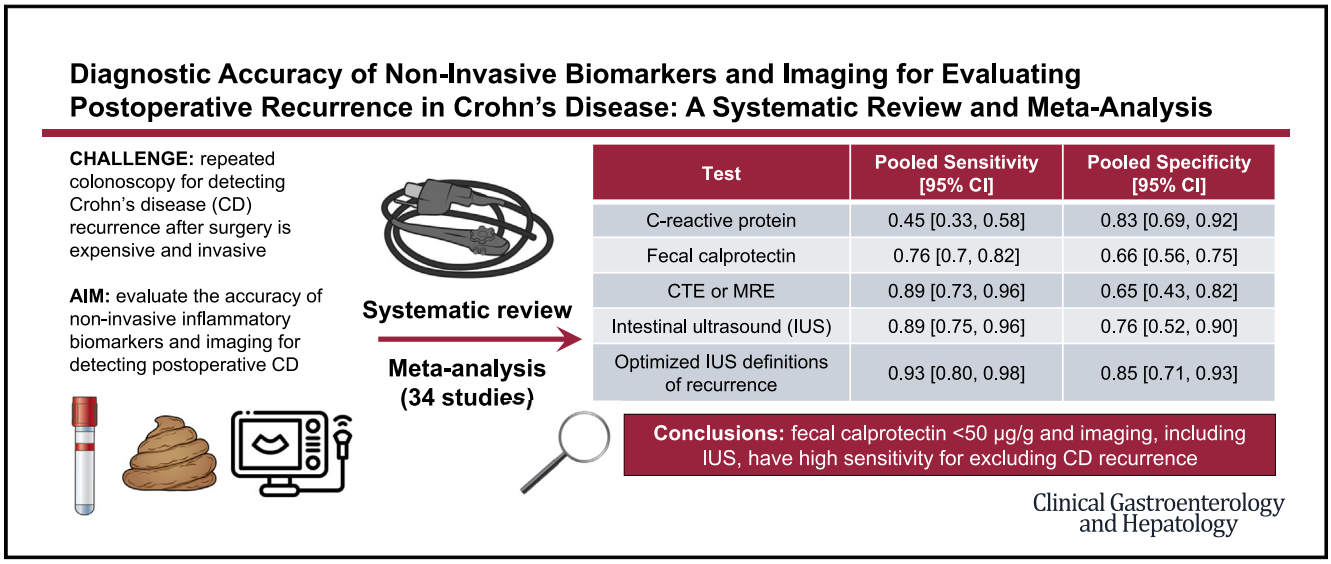
Diagnostic Accuracy of Noninvasive Biomarkers and Imaging for Evaluating Postoperative Recurrence in Crohn's Disease

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BACKGROUND & AIMS: **Colonoscopy is recommended to monitor for Crohn's disease (CD) recurrence after surgical resection. However, repeated colonoscopy is invasive and resource-intensive. We conducted a systematic review and meta-analysis to evaluate the pooled diagnostic accuracy of noninvasive biomarkers and imaging measures for detecting endoscopic CD recurrence, as compared with colonoscopy.**

METHODS: **A systematic review was conducted to January 31, 2024, to identify studies evaluating the diagnostic accuracy of C-reactive protein (CRP), fecal calprotectin, computed tomography and magnetic resonance enterography, or intestinal ultrasound (IUS) compared with colonoscopy for detecting CD recurrence. Estimates of sensitivity, specificity, and positive and negative**

Abbreviations used in this paper: AUC, area under the curve; BWT, bowel wall thickness; CD, Crohn's disease; CI, confidence interval; CRP, C-reactive protein; CTE, computed tomography enterography; DOR, diagnostic odds ratio; HSROC, hierarchical summary receiver operating characteristic; IUS, intestinal ultrasound; LR, likelihood ratio; MRE, magnetic resonance enterography; RCT, randomized controlled trial; ROC, receiver operating characteristic; RS, Rutgeerts score.

Most current article

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1542-3565
https://doi.org/10.1016/j.cgh.2025.03.030

likelihood ratios were pooled using a random-effects hierarchical summary receiver operating characteristic model.

RESULTS: A total of 17 studies (N = 1080) evaluated inflammatory biomarkers and 20 studies (N = 1053) assessed imaging measures. The pooled sensitivity and specificity of CRP (threshold, 5.0 mg/L) were 0.45 (95% confidence interval [CI], 0.33–0.58) and 0.83 (95% CI, 0.68–0.92), respectively. Fecal calprotectin (threshold, 50 μg/g) was moderately sensitive 0.76 (95% CI, 0.70–0.82) but less specific 0.66 (95% CI, 0.56–0.75). Sensitivity for computed tomography enterography/magnetic resonance enterography and IUS was 0.89 (95% CI, 0.73–0.96) and 0.92 (95% CI, 0.75–0.96); specificity was 0.65 (95% CI, 0.43–0.82) and 0.76 (95% CI, 0.52–0.90), respectively. Using optimized radiographic parameters for IUS, specificity was improved to 0.85 (95% CI, 0.71–0.93).

CONCLUSIONS: The high sensitivity of fecal calprotectin (<50 μg/g) and cross-sectional imaging can help reduce the need for invasive and costly colonoscopy monitoring for CD recurrence after surgery. Applying optimal definitions of sonographic recurrence by IUS parameters may further improve specificity for making therapeutic decisions without endoscopy.

Keywords: Biomarkers; Colonoscopy; Crohn's Disease; Imaging; Monitoring; Recurrence; Ultrasound.

Despite substantial advancements in the medical management of moderately-to-severely active Crohn's disease (CD), surgical resection is required in up to half of patients with refractory or complicated disease.[1] Surgery for CD is not curative, and many patients experience postoperative recurrence.[2] Current clinical practice guidelines recommend the use of prophylactic medical therapy for patients at high risk for recurrence, which has been categorized using nonspecific characteristics such as younger age at diagnosis, active smoking, multiple prior resections, or penetrating disease.[3] Endoscopic monitoring by ileocolonoscopy is recommended, regardless of risk factors, for postoperative recurrence. The value of endoscopic surveillance was demonstrated in the POCER trial, where use of a risk-stratified medical treatment approach combined with early colonoscopy at 6 months after surgery was superior to standard care for reducing the rate of 18-month endoscopic recurrence (49% vs 67%; $P = .03$).[4] Accurate endoscopic assessment of the surgical anastomosis and neoterminal ileum can help direct early and appropriate clinical decisions around pharmacologic therapy, particularly because the symptoms of CD may often confounded after surgery by concomitant bile acid diarrhea, short gut, or small intestinal bacterial overgrowth.[5]

Although ileocolonoscopy is the gold standard for detecting postoperative CD recurrence, endoscopy is invasive, resource-intensive, requires bowel preparation and sedation, and can result in patient discomfort or procedural complications. Accordingly, there has been increasing interest in the use of noninvasive modalities for assessing CD after surgery, including the use of inflammatory biomarkers such as fecal calprotectin and C-reactive protein (CRP), and bowel imaging using computed tomography (CTE) or magnetic resonance enterography (MRE) and intestinal ultrasound (IUS). Targeting tight disease control using fecal calprotectin and CRP to monitor and intervene early on subclinical disease activity was demonstrated to be superior to standard care in the CALM trial, and increasingly, IUS is being used to noninvasively monitor CD at the point of care and make treatment changes when transmural remission or response has not been achieved.[6] However, these applications have been primarily based on data from patients with luminal CD, rather than in the postoperative setting. Specifically understanding the operating characteristics of noninvasive biomarkers and imaging for accurately detecting postoperative CD recurrence will help guide clinicians and patients in ordering appropriate investigations, interpreting these results, and initiating timely treatment.

In this systematic review and meta-analysis, we assessed the diagnostic accuracy of fecal calprotectin, CRP, CTE, MRE, and IUS compared with ileocolonoscopy as the gold standard for detecting CD recurrence.

Methods

This meta-analysis was conducted and reported using the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) standards and the extension for diagnostic test accuracy.[7]

Study Eligibility

We included all studies that reported the diagnostic test accuracy for fecal calprotectin, CRP, CTE, MRE, or IUS for detecting CD recurrence, as compared with colonoscopy as the gold standard. Endoscopic recurrence was defined by a Rutgeerts' score (RS) ≥i2 (or i2b in studies using the modified RS), as defined by ileocolonoscopy; studies using capsule endoscopy as the reference standard were not included. Studies were considered eligible

if they reported these parameters among adult patients ≥18 years undergoing ileocolonic resection with primary anastomosis. Included studies reported the sensitivity, specificity, and positive and negative likelihood ratio (LR+, LR−) of noninvasive testing measures, or reported the parameters required to calculate these metrics. We included randomized controlled trials (RCTs), prospective and retrospective observational cohort studies, and case-control studies.

Data Sources, Search Strategy, and Study Selection

MEDLINE (Ovid) and Embase (Ovid) were searched from inception to January 31, 2024, using search terms to capture CD, surgery, and postoperative recurrence (Supplemental Appendix 1). The search was designed and executed in consultation with a medical librarian and specifically included search terms to capture multiple permutations of key words, in both text, titles and abstracts, and keyword headings. Two authors (SS, CR) independently screened the titles and abstracts to identify potentially relevant studies. Full text review was then conducted to confirm study eligibility. Secondary screening from the reference lists of the included articles and pertinent reviews was conducted to identify a comprehensive list of other potentially eligible studies. Data were extracted in duplicate and independently by 2 authors, with any discrepancies adjudicated by the senior author (CM).

Data Collection and Assessment of Study Quality

All data were collected using a standardized data extraction sheet. The following covariables were collected: study author, year of publication, journal, study design, duration of follow-up, location of study, definition of CD recurrence, study intervention(s) and comparators, and diagnostic test accuracy (sensitivity, specificity, positive and negative LR; or the corresponding number of true and false positive and negative cases for calculating diagnostic parameters; studies only reporting area under the curve [AUC] or where diagnostic parameters could not be calculated were excluded). Study quality for diagnostic accuracy was assessed using the QUADAS-2 tool.[8]

Statistical Methods

The primary outcome was the diagnostic accuracy of noninvasive testing compared with ileocolonoscopy for detecting recurrence. Different methods for pooling diagnostic accuracy have previously been used, including bivariate and hierarchical summary receiver operating characteristic (HSROC) models. For the primary analysis, we conducted diagnostic test accuracy meta-analysis using the HSROC model, which is preferred when different studies report different testing thresholds.[9–11] The HSROC model was implemented using the Cochrane Collaboration MetaDAS macro, which is based on the PROC NLMIXED procedure in SAS. To ensure robustness of our findings, we also estimated diagnostic testing accuracy in Stata using the *metadta* package and evaluated bivariate random- and fixed-effects models to ensure that the estimates of accuracy were stable. Diagnostic accuracy is expressed using the generalized sensitivity, specificity, positive and negative LR (LR+/LR−), diagnostic odds ratio (DOR), and summary receiver under the receiver operating characteristic (ROC) curves, with corresponding I^2 measures of heterogeneity.[12]

Statistical analyses were performed in SAS (version 9.4, SAS Institute) and Stata (version 18.0, StataCorp LLC).

What You Need to Know

Background

Colonoscopy is recommended for surveillance of Crohn's disease (CD) recurrence after surgery. However, repeated colonoscopy is invasive and resource-intensive. This study evaluated the diagnostic accuracy of noninvasive tools for detecting endoscopic recurrence.

Findings

Fecal calprotectin <50 μg/g and normal imaging by computed tomography enterography, magnetic resonance enterography, or intestinal ultrasound (IUS) have high sensitivity for excluding active CD recurrence. Using optimal definitions of sonographic recurrence minimizes the occurrence of false-positive results.

Implications for patient care

The use of noninvasive biomarkers and abdominal imaging can potentially reduce the need for frequent, invasive colonoscopies, which has implications for health care resource utilization and patient management paradigms after surgery for CD.

Results

Included Studies

The search strategy identified 5009 citations. Following removal of duplicates and screening titles and abstracts, 221 studies underwent full-text review, and 34 studies were included in quantitative analysis (Supplemental Figure 1).[13–37] A total of 17 studies evaluated CRP and/or fecal calprotectin for detecting CD recurrence (Table 1), including 1080 patients. Testing

Table 1. Characteristics of Studies Evaluating Fecal Calprotectin and CRP for Detecting Postoperative Crohn's Disease Recurrence

Author (year)	Study location	Study design	Definition of endoscopic recurrence	Prevalence of recurrence, % (n/N)	Diagnostic test	Test threshold	Sensitivity	Specificity	PPV	NPV
Orlando (2006)	Italy	Prospective, observational 1 year follow-up	RS ≥i2	48.7% (19/39)	Fecal calprotectin	50 ug/g	0.89	0.25	0.53	0.71
						100 ug/g	0.79	0.45	0.58	0.69
						150 ug/g	0.79	0.55	0.63	0.73
						200 ug/g	0.63	0.75	0.71	0.68
						250 ug/g	0.53	0.85	0.75	0.68
Regueiro (2011)	U.S.	Prospective, observational 1 year follow-up	RS ≥i2	50.0% (12/24)	CRP	5 mg/L	0.50	0.50	0.50	0.50
						10 mg/L	0.25	0.83	0.60	0.53
Lobaton (2013)	Spain	Prospective, observational 1 year follow-up	CDEIS ≥3	40.4% (36/89)	Fecal calprotectin	272 ug/g	0.97	0.79	0.76	0.98
						274 ug/g	0.97	0.77	0.75	0.98
Lasson (2014)	Sweden	Prospective, observational 1 year follow-up	RS ≥i2	43.3% (13/30)	Fecal calprotectin	100 ug/g	0.85	0.35	0.50	0.75
						200 ug/g	0.69	0.53	0.53	0.69
						250 ug/g	0.46	0.53	0.43	0.56
Wright (2015)	Australia	Prospective, observational 6–18 month follow-up	RS ≥i2	34.8% (24/69)	Fecal calprotectin	50 ug/g	0.96	0.36	0.44	0.94
						100 ug/g	0.88	0.58	0.53	0.90
						127 ug/g	0.88	0.67	0.58	0.91
						150 ug/g	0.79	0.69	0.58	0.86
						200 ug/g	0.71	0.71	0.57	0.82
Boschetti (2015a)	France	Prospective, observational 18 month follow-up	RS ≥i2	50.0% (43/86)	Fecal calprotectin	50 ug/g	0.98	0.33	0.59	0.93
						100 ug/g	0.95	0.53	0.67	0.92
						150 ug/g	0.77	0.81	0.80	0.78
						250 ug/g	0.51	0.91	0.85	0.65
					CRP	1 mg/L	0.86	0.23	0.53	0.63
						2 mg/L	0.72	0.65	0.67	0.70
						5 mg/L	0.42	0.86	0.75	0.60
Boschetti (2015b)	France	Prospective, observational 6–12 month follow-up	RS ≥i2	32.4% (12/37)	CRP	5 mg/L	0.42	0.88	0.63	0.76
Lopes (2016)	Portugal	Prospective, observational Median 25 months from surgery	RS ≥i2	75.8% (75/99)	Fecal calprotectin	50 ug/g	0.72	0.71	0.89	0.45
						100 ug/g	0.48	0.79	0.88	0.33
			Modified RS ≥i2b	34.3% (34/99)		50 ug/g	0.94	0.55	0.52	0.95
						100 ug/g	0.74	0.75	0.61	0.84
Bachiller (2016)	Spain	Retrospective, observational Follow-up not reported	Modified RS ≥i2b	58.8% (57/97)	Fecal calprotectin	50 ug/g	0.88	0.43	0.68	0.71
						60 ug/g	0.88	0.58	0.75	0.77
						100 ug/g	0.75	0.53	0.69	0.60
						150 ug/g	0.51	0.63	0.66	0.47
						200 ug/g	0.35	0.73	0.65	0.44

Table 1. Continued

Author (year)	Study location	Study design	Definition of endoscopic recurrence	Prevalence of recurrence, % (n/N)	Diagnostic test	Test threshold	Sensitivity	Specificity	PPV	NPV
Garcia-Planella (2016)	Spain	Prospective, observational Median 63.9 months from surgery	RS ≥i2	41.2% (49/119)	Fecal calprotectin	50 ug/g 100 ug/g	0.86 0.69	0.27 0.64	0.45 0.58	0.73 0.75
Yamamoto (2016)	Japan	Prospective, observational 24-month follow-up	RS ≥i2	26.7% (8/30)	Fecal calprotectin	140 ug/g	0.75	0.91	0.75	0.91
Baillet (2018)	France	Prospective, observational 1-year follow-up	Modified RS ≥i2b	43.3% (13/30)	Fecal calprotectin CRP	100 ug/g 5 mg/L	0.69 0.31	0.94 0.88	0.90 0.67	0.80 0.63
Verdejo (2018)	Spain	Cross-sectional, observational Median 4.5 years from surgery	RS ≥i2	57.0% (49/86)	Fecal calprotectin	50 ug/g 62 ug/g 100 ug/g 200 ug/g 250 ug/g	0.90 0.86 0.67 0.47 0.43	0.41 0.46 0.57 0.78 0.89	0.67 0.68 0.67 0.74 0.84	0.75 0.71 0.57 0.53 0.54
					CRP	0.32 mg/L 0.5 mg/L 0.75 mg/L 1 mg/L 1.5 mg/L	0.57 0.41 0.31 0.22 0.14	0.86 0.92 0.92 0.95 1.00	0.85 0.87 0.83 0.85 1.00	0.60 0.54 0.50 0.48 0.47
Boube (2020)	France	Prospective, observational 6-month follow-up	Modified RS ≥i2b	37.5% (18/48)	Fecal calprotectin	76 ug/g 102 ug/g	0.61 0.44	0.37 0.77	0.37 0.53	0.61 0.70
Veyre (2021)	France	Prospective, observational 6–12 months follow-up	RS ≥i2	41.8% (23/55)	Fecal calprotectin	50 ug/g 65 ug/g 75 ug/g 105 ug/g	0.96 0.96 0.87 0.78	0.19 0.31 0.34 0.53	0.46 0.50 0.49 0.55	0.86 0.91 0.79 0.77
Daoud (2022)	U.S.	Retrospective, observational Follow-up not reported	Ileocolonoscopy	27.5% (14/51)	Fecal calprotectin	60 ug/g 120 ug/g 200 ug/g	0.88 0.69 0.50	0.91 0.94 0.94	0.82 0.85 0.80	0.94 0.87 0.80
Furfaro (2023)	Italy	Prospective, observational 1-year follow-up	RS ≥i2	65.9% (60/91)	Fecal calprotectin	50 ug/g	0.83	0.65	0.82	0.67

CRP, C-reactive protein; NPV, negative predictive value; PPV, positive predictive value; RS, Rutgeerts score; U.S., United States.

thresholds for fecal calprotectin ranged from 50 μg/g to 274 μg/g and 0.32 mg/L to 10 mg/L for CRP. A total of 20 studies evaluated imaging modalities for detecting CD recurrence (Table 2), including 1053 patients. Most studies (14/17 evaluating biomarkers and 16/20 evaluating imaging) were prospective in nature, and most were deemed to be at low risk for bias (Supplemental Appendix 2).

Diagnostic Accuracy of Inflammatory Biomarkers for Postoperative Recurrence

A total of 5 unique studies describing the diagnostic accuracy at multiple different thresholds for CRP concentrations and 15 studies describing the diagnostic accuracy of fecal calprotectin were included in pooled analysis. Summary measures of sensitivity, specificity, +LR/−LR, and DOR are demonstrated in Table 3. For both measures, an HSROC model was fit (Figure 1), and a random-effects model showed better fit than a fixed-effects bivariate model (LR test $P < .0001$ for both CRP and fecal calprotectin).

For CRP, the testing threshold of the HSROC model was centered at 5.0 mg/L, with pooled sensitivity of 0.45 (95% confidence interval [CI], 0.33–0.58) and specificity of 0.83 (95% CI, 0.68–0.92). There was no significant heterogeneity for the generalized model ($I^2 = 0\%$). The CRP test threshold of 5.0 mg/L had a statistically significant impact on sensitivity, specificity, and relative DOR: for increasing thresholds of CRP, the sensitivity decreases, specificity increases, and relative DOR increases (all $P < .001$). For fecal calprotectin, the test threshold was centered at 50 ug/g (scaled by 10-ug/g increments) with pooled sensitivity of 0.76 (95% CI, 0.70–0.82) and specificity of 0.66 (95% CI, 0.56–0.75). Increasing this threshold significantly improved specificity with moderate reductions in sensitivity (for every 10-μg/g increase in fecal calprotectin threshold, the LR+ increased by 2.25 (95% CI, 1.71–2.97), whereas the LR− increases by 0.30 (95% CI, 0.19–0.46). There was no significant heterogeneity for the fecal calprotectin generalized model ($I^2 = 43\%$).

Diagnostic Accuracy of Imaging for Postoperative Recurrence

A total of 7 studies evaluated the role of CTE and/or MRE, using 11 different definitions of recurrence (Table 2). Pooled sensitivity and specificity for CTE or MRE was 0.89 (95% CI, 0.73–0.96) and 0.65 (95% CI, 0.43–0.82), respectively (Figure 2, Table 3). Similar findings were demonstrated in a sensitivity analysis using MRE alone (sensitivity, 0.92; 95% CI, 0.35–1.00; specificity, 0.73; 95% CI, 0.49–0.89), which reduced heterogeneity in the model ($I^2 =$ 54% for combined CTE and MRE; $I^2 = 31\%$ for MRE alone).

A total of 13 studies evaluated the accuracy of ultrasound imaging, using a total of 29 different definitions of sonographic recurrence of CD after surgery, with or without contrast enhancement. The most used parameters for defining sonographic recurrence were based on bowel wall thickness (BWT) >3 to 5 mm at the neoterminal ileum, color Doppler flow, inflammatory fat, and in some studies, inclusion of complications such as development of stenosis, fistula, or abscess. Despite differing IUS-based definitions of recurrence, there was no statistically significant heterogeneity in the generalized model ($I^2 = 26\%$). Pooled estimates of sensitivity was 0.89 (95% CI, 0.75–0.96) and specificity was 0.76 (95% CI, 0.52–0.90) on IUS. These were further improved in a sensitivity analysis when analyzing only the best performing definition of IUS recurrence (sensitivity, 0.93; 95% CI, 0.80–0.98; specificity, 0.85; 95% CI, 0.71–0.93), which included methods such as adding oral contrast, increasing the threshold for BWT to 5 mm from 3 mm, or combining multiple sonographic parameters including abnormal contrast enhancement, Doppler flow, or mesenteric lymph nodes (Table 2).

Discussion

The recurrence of CD after surgery is a common clinical challenge and is associated with increased risks of clinical relapse, disease progression, and requirement for second surgery. Although current clinical practice guidelines recommend a risk-stratified medical prophylaxis approach and repeated colonoscopy for evaluating disease recurrence, this approach may not be feasible or preferable for all patients based on invasiveness, cost, and health care utilization. In this systematic review and meta-analysis, we evaluated the diagnostic performance of noninvasive methods for detecting CD recurrence after surgery. We highlight several clinically important findings. First, we show that CRP is relatively specific but inadequately sensitive to be used in isolation for monitoring of recurrence. In contrast, at very low thresholds (<50 μg/g), fecal calprotectin is more sensitive but lacks specificity. Second, imaging modalities are highly sensitive measures for recurrence. Optimizing the radiographic parameters used to define recurrence, particularly with IUS, may be a promising avenue forwards as this can improve the specificity of sonographic diagnosis of recurrence without the need for repeated colonoscopy. These findings are highly relevant to clinicians when deciding which test(s) to order after surgery.

Although the value of treating to CRP and fecal calprotectin targets using a tight-control therapeutic strategy has been demonstrated in luminal CD,[6] these findings may not directly translate to the postoperative setting. CRP is insufficiently sensitive for detecting recurrence, particularly as patients with isolated disease at the anastomosis or distal neoterminal ileum may not generate a robust systemic immune response.[38] Fecal calprotectin appears to be a more sensitive measure of recurrence: in a post-hoc analysis of the POCER trial,

Table 2. Characteristics of Studies Evaluating Bowel Imaging for Detecting Postoperative Crohn's Disease Recurrence

Author (year)	Study location	Study design	Prevalence of endoscopic recurrence, % (n/N)	Diagnostic test	Test threshold	Sensitivity	Specificity	PPV	NPV
Andreoli (1998)	Italy	Prospective, observational Mean 35.4 months from surgery	78.0% (32/41)	IUS	BWT >5 mm	0.81	0.89	0.96	0.57
Rispo (2006)	Italy	Prospective, observational 1-year follow-up	53.3% (24/45)	IUS	BWT >3 mm BWT >5 mm	0.79 0.94	0.95 1.00	0.95 1.00	0.80 0.97
Orlando (2006)	Italy	Prospective, observational 1-year follow-up	48.7% (19/39)	IUS	BWT ≥5 mm	0.26	0.90	0.71	0.56
Biancone (2007)	Italy	Prospective, observational 1-year follow-up	95.5% (21/22)	SICUS	Median BWT >5 mm	1.00	0.00	0.95	0.00
Castiglione (2008)	Italy	Prospective, observational	55.0% (22/40)	IUS	BWT >3 mm, bowel distention, stricture, fistulae, or abscess	0.77	0.94	0.94	0.77
		1-year follow-up		IUS with oral contrast	BWT >3 mm, bowel distention, stricture, fistulae, or abscess	0.82	0.94	0.95	0.81
Calabrese (2009)	Italy	Prospective, observational 6-month follow-up	93.1% (67/72)	IUS	Small bowel dilation (luminal diameter >2.5 cm), stricture (lumen diameter <1 cm), fistulae (hypoechoic tract with or without hyperechoic content, mesenteric adipose tissue alteration and/or masses, abscess (roundish anechoic lesion(s) with irregular wall and often with internal echoes and posterior echo enhancement)	0.93	0.20	0.94	0.17
Onali (2010)	Italy	Prospective, observational 36-month follow-up	96.0% (24/25)	SICUS	BWT >3 mm, stiff loop (increased BWT not distended by contrast solution, small bowel dilation >2.5 cm, stricture (lumen diameter <1 cm), fistulae (hypoechoic tract with or without hyperechoic content,	1.00	0.00	0.96	0.00

Table 2. Continued

Author (year)	Study location	Study design	Prevalence of endoscopic recurrence, % (n/N)	Diagnostic test	Test threshold	Sensitivity	Specificity	PPV	NPV
					mesenteric enlargement and/or masses, abscess (roundish anechoic lesion(s) with irregular wall and often with internal echoes and posterior echo enhancement)				
Paredes (2010)	Spain	Prospective, observational Mean 87 months from surgery	65.0% (26/40)	US with Tc scintigraphy	BWT of neoterminal ileum >3 mm and/or color Doppler flow positive BWT of neoterminal ileum >5 mm Color Doppler flow grade 2 or 3	0.77 0.80 0.80	0.57 0.78 0.67	0.87 0.75 0.67	0.40 0.82 0.80
Gallego Ojea (2010)	Spain	Prospective, observational Median 58 months from surgery	83.3% (25/30)	MRE	BWT, relative enhancement, motility, stenosis, intestinal edema, mucosal abnormalities, lymph nodes, fistula or inflammatory mass	1.00	0.60	0.93	1.00
Aguas (2012)	Spain	Prospective, observational 1-year follow-up	20.7% (6/29)	MRE	Distinct BWT, mural contrast enhancement, low grade stenosis without prestenotic dilation or high-grade stenosis, transmural edema with T2W signal increase and contrast enhancement of perienteric fat, extramural complications (fistula, abscess, conglomeration of bowel loops)	1.00	0.92	0.86	1.00
Mao (2013)	China	Prospective, observational 30-month follow-up	75.0% (24/32)	CTE	CTE1 (minor mucosal irregularities with sligh wall thickening and mural contrast enhancement), CTE2 (mucosal hyperdensity with distinct	0.96	0.88	0.96	0.88

Table 2. Continued

Author (year)	Study location	Study design	Prevalence of endoscopic recurrence, % (n/N)	Diagnostic test	Test threshold	Sensitivity	Specificity	PPV	NPV
					BWT, no stenosis, or stenosis without prestenotic dilation), CTE3 (major mucosal abnormalities, distinct BWT with target sign and extravisceral signs such as perienteric stranding, comb sign, fibrofatty proliferation, stenosis with prestenotic dilation, and/or the presence of complications)				
Paredes (2013)	Spain	Prospective, observational Median 60 months from surgery	81.7% (49/60)	IUS/CEUS	BWT of neoterminal ileum >3 mm and/or colour Doppler flow positive Contrast enhancement of neoterminal ileum >34.5%	0.90 0.98	0.82 0.82	0.96 0.96	0.64 0.90
Onali (2016)	Italy	Prospective, observational 5-year follow-up	97.5% (39/40)	IUS	BWT ≥3 mm	1.00	0.00	1.00	0.00
Martinez (2018)	Spain	Prospective, observational Mean 6 years from surgery	83.3% (90/108)	IUS/CEUS	BWT ≥3 mm Doppler color ≥ grade 1 Wall contrast enhancement ≥46%	0.94 0.82 0.89	0.72 0.67 0.83	0.94 0.93 0.96	0.72 0.43 0.60
Baillet (2018)	France	Prospective, observational Median 6 months from surgery	43.3% (13/30)	MR	Edema, ulcers, bowel wall thickening (mm), adjacent enlarged lymph nodes (>8 mm in shortest diameter), comb sign, penetrating complications (fistula, abscess, phlegmon), and contrast enhancement	0.54	0.82	0.70	0.70
Carmona (2022)	Spain	Retrospective, observational Follow-up not reported	51.6% (16/31)	IUS	BWT >3.4 mm Bowel wall hyperemia Loss of layered pattern Presence of inflammatory fat	1.00 0.93 0.64 0.44	0.87 0.87 1.00 1.00	0.89 0.88 1.00 1.00	1.00 0.93 0.74 0.63

Table 2. Continued

Author (year)	Study location	Study design	Prevalence of endoscopic recurrence, % (n/N)	Diagnostic test	Test threshold	Sensitivity	Specificity	PPV	NPV
Bachour (2022)	U.S.	Retrospective, observational Median 4.5-year follow-up	38.9% (84/216)	CTE/MRE	Intestinal wall thickening, luminal narrowing, hyperenhancement, mural fat, upstream dilation and stasis	0.89	0.32	0.45	0.82
Pozzessere (2022)	Italy	Retrospective, observational Mean 8 years from surgery	71.0% (44/62)	MRE	BWT >3 mm at anastomosis	0.82	0.89	0.95	0.67
					Mucosal wall enhancement	0.63	0.06	0.66	0.06
					Presence of pathologic lymph nodes	0.65	0.18	0.59	0.22
					Segmental wall enhancement	0.82	0.89	0.95	0.67
					Hyperintensity on diffusion weighted imaging	0.70	0.40	0.76	0.33
Furfaro (2023)	Italy	Prospective, observational 1-year follow-up	65.9% (60/91)	IUS	BWT ≥3 mm	0.77	0.65	0.81	0.59
					Presence of pathologic lymph nodes	0.35	0.97	0.95	0.43
					BWT ≥3 mm and fecal calprotectin ≥50 ug/g	0.65	0.93	0.94	0.59
					BWT ≥3 mm, fecal calprotectin ≥50 ug/g, pathologic lymph nodes	0.33	1.00	1.00	0.60

BWT, bowel wall thickness; CT, computed tomography; CTE, computed tomography enterography; IUS, intestinal ultrasound; MR, magnetic resonance; MRE magnetic resonance enterography; NPV, negative predictive value; PPV, positive predictive value; SICUS, small intestine contrast ultrasonography; US, ultrasound; U.S., United States.

Table 3. Diagnostic Accuracy of CRP, Fecal Calprotectin, and Bowel Imaging for Detecting Endoscopic Recurrence of Crohn's Disease

Test	Pooled sensitivity (95% CI)	Pooled specificity (95% CI)	Pooled LR+ (95% CI)	Pooled LR− (95% CI)	Pooled DOR (95% CI)
CRP	0.45 (0.33–0.58)	0.83 (0.69–0.92)	2.67 (1.69–4.23)	0.66 (0.57–0.75)	4.04 (2.55–6.44)
Fecal calprotectin	0.76 (0.7–0.82)	0.66 (0.56–0.75)	2.27 (1.68–3.05)	0.35 (0.26–0.48)	6.38 (3.7–11.01)
CTE or MRE	0.89 (0.73–0.96)	0.65 (0.43–0.82)	2.53 (1.37–4.66)	0.17 (0.06–0.51)	15.11 (3.12–73.25)
IUS	0.89 (0.75–0.96)	0.76 (0.52–0.90)	3.71 (1.78–7.78)	0.14 (0.07–0.3)	26.1 (11.82–57.63)

CI, confidence interval; CRP, C-reactive protein; CTE, computed tomography enterography; DOR, diagnostic odds ratio; IUS, intestinal ultrasound; LR, likelihood ratio; MRE, magnetic resonance enterography.

Wright et al demonstrated moderate correlations between fecal calprotectin concentrations and the presence and severity of endoscopic lesions, although the sensitivity and specificity of the test depend on the testing threshold.[35] At lower thresholds of 50 to 100 ug/g, fecal calprotectin has high negative predictive value and may allow deferral of colonoscopy, although confidence in treating and monitoring therapeutic response using fecal calprotectin alone may be diminished due to poor specificity. Furthermore, it should be noted that there is substantial inter- and intra-individual variability in fecal calprotectin concentrations, as well as variations between testing assays.[39]

In our pooled analysis, imaging by cross-sectional modalities or IUS was highly sensitive for disease recurrence, although specificity varied. Two hypotheses may explain these findings. First, it is possible that radiographic modalities capture relevant disease-related findings that are not visualized luminally on colonoscopy. Indeed, the concept of "mucosal healing" in CD has already begun to evolve from endoscopic healing to also capturing transmural remission that requires cross-sectional evaluation.[40] Second, low specificity may result from the application of "normal" parameters from native bowel to the post-surgical setting. For instance, while most IUS studies used a BWT threshold of $\geq$3 mm to define postoperative recurrence, this cutoff is appropriate for normal bowel, and it is unknown whether the same threshold should be applied after surgery. It is also unclear how these parameters should be scored at the anastomosis and neoterminal ileum, and whether this should differ depending on the type of anastomosis or surgery (particularly for side-to-side and anti-peristaltic anastomoses).[41] To improve specificity, some authors have proposed using BWT $\geq$5 mm as the threshold in the post-surgical setting or combining multiple measures such as color Doppler flow or presence of inflammatory lymph nodes. In our sensitivity analysis, using the best performing definitions of postoperative CD recurrence on IUS, both sensitivity and specificity could be improved, highlighting the need for an evidence-based consensus for how to assess postoperative CD recurrence.

Our findings also highlight that, although we pooled diagnostic accuracy across different studies evaluating IUS, there is substantial heterogeneity in how ultrasound is applied in routine care and the operator-dependent nature of this modality. For example, there may be differences between centers in how fasting is performed prior to IUS, the application of small intestinal contrast,

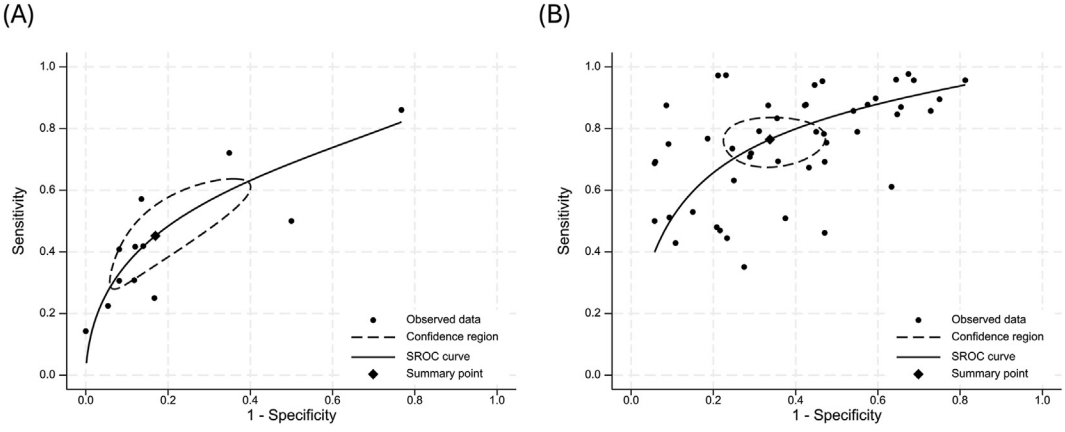


Figure 1. HSROC models for diagnostic accuracy of CRP (*A*) and fecal calprotectin (*B*) for detecting endoscopic recurrence of Crohn's disease.

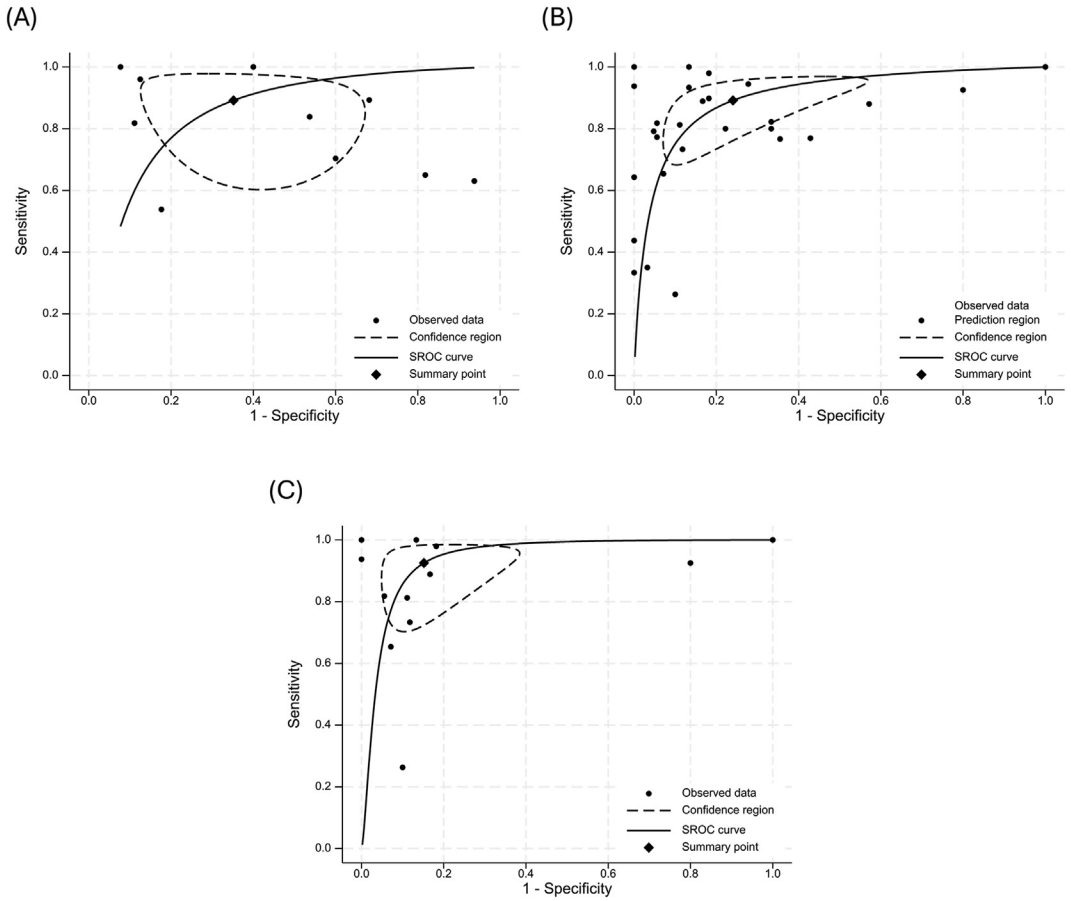


Figure 2. HSROC models for diagnostic accuracy and pooled sensitivity and specificity of CTE or MRE (*A*), IUS (*B*), and IUS with optimized definitions (*C*) for detecting endoscopic recurrence of Crohn's disease.

the use of different equipment ranging from high specification machines to handheld point-of-care devices, and the training, experience, and interpretation based on the operator themselves. Although there was no statistical heterogeneity observed in our findings for IUS, there may be important clinical differences in how this technology is applied and interpreted. For example, a recent study by Goodsall et al identified that there may be substantial differences in interpretation of IUS parameters, even when assessed by highly trained experts in the modality.[42]

Another possible approach to maximize the positive and negative predictive value is to combine multiple modalities of noninvasive testing. For example, Furfaro et al evaluated 91 consecutive patients with CD in 3 Italian referral centers undergoing ileocecal resection.[20] Although BWT was strongly associated with endoscopic recurrence, the combination of both BWT $\geq$3 mm and fecal calprotectin $\geq$50 μg/g correctly classified 75% of patients with endoscopic recurrence. A risk-stratified approach to selecting diagnostic testing modalities may also be warranted. For example, in patients with a low pre-test probability of recurrence, combining 2 highly sensitive measures (eg, fecal calprotectin $<$50 μg/g and IUS with BWT $<$3 mm) may optimally rule out recurrence without colonoscopy. Conversely, for patients with a high likelihood for recurrence, combining 2 highly specific measures (eg, CRP $>$5 mg/L, fecal calprotectin $>$250 μg/g, IUS with BWT $>$5 mm, or multiple sonographic features of inflammation) may give clinicians sufficient confidence to start or change pharmacologic treatment. Recently, the American Gastroenterological Association published guidelines on the appropriate use of biomarkers in CD: in the postoperative setting for asymptomatic patients at low risk for recurrence or who are already on pharmacologic prophylaxis, a fecal calprotectin $<$50 μg/g was proposed as a threshold with high sensitivity allowing for deferral of routine endoscopic evaluation.[43] Although we agree with these guidelines, we propose that IUS, as a noninvasive, repeatable, low-cost, high sensitivity diagnostic tool, can be added to give confidence in deferring endoscopy in settings where there is an indeterminate risk for recurrence (eg, fecal calprotectin $>$50 μg/g but $<$250 μg/g) (Figure 3). In settings where there is a higher risk for recurrence (eg, elevated fecal calprotectin or CRP, which has high specificity), we consider endoscopy a necessary investigation to evaluate the severity of recurrence and establish a new baseline for assessing response to subsequent therapy.

Our study has several strengths. First, we conducted a comprehensive systematic review and used appropriate

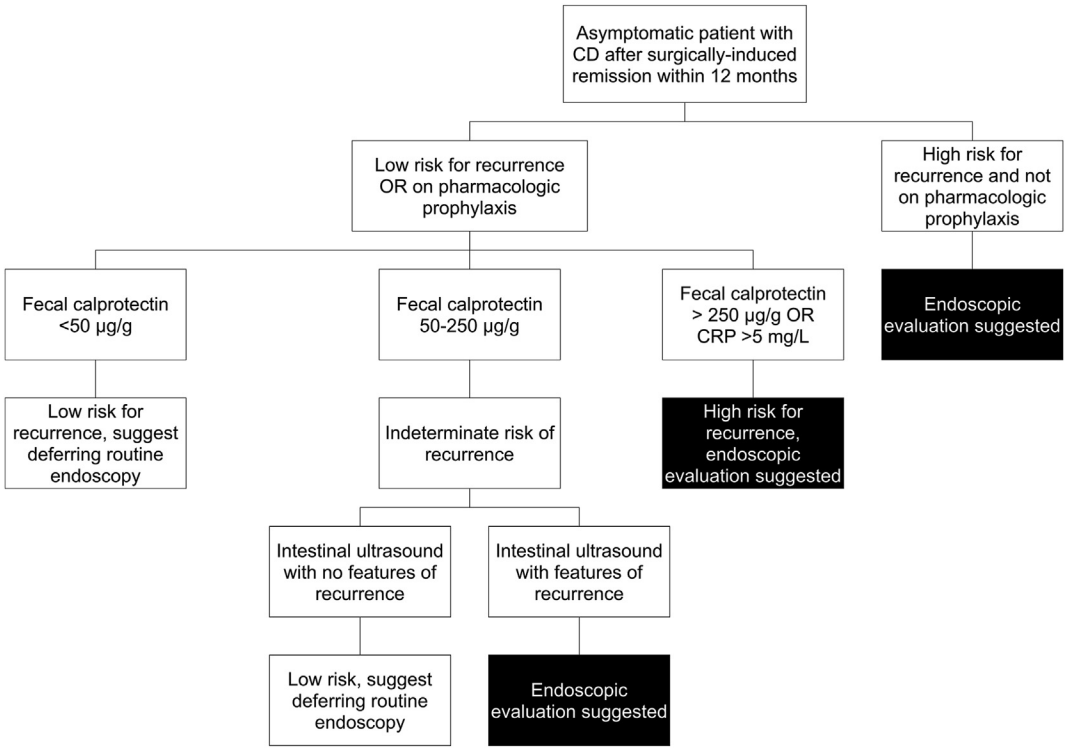


Figure 3. Proposed clinical algorithm for noninvasive assessment of postoperative Crohn's disease recurrence.

methods to pool exposure-adjusted incidence rates for recurrence, assessed the recurrence rate across different clinical scenarios, and used rigorous methods to evaluate the diagnostic accuracy of commonly used noninvasive tests. However, we recognize that our study also has some important limitations. First, although there was no statistically significant heterogeneity, there were differences in the designs and populations of patients included in our analysis. Second, most studies defined endoscopic recurrence based on the RS. Although this is the most used and accepted index for postoperative CD, it is a prognostic rather than evaluative tool, there are limitations with respect to interpreting the different grades of the RS (for which the descriptors of each grade are not mutually exclusive), and there has been demonstrated inter-rater disagreement on the RS scoring.[44] Third, although diagnostic accuracy parameters such as sensitivity and specificity are often thought to be prevalence-independent measures, this is overly simplistic, as previous analyses have demonstrated overestimation of sensitivity and underestimation of specificity with increasing disease prevalence.[45] There were insufficient data to estimate prevalence-specific estimates of diagnostic accuracy in this analysis. Fourth, follow-up durations were highly variable across studies, which made it challenging for us to validly meta-regress the effect of duration from surgery on diagnostic accuracy. Furthermore, our findings are generalizable to the asymptomatic patient after surgery; very few studies in our analysis evaluated the diagnostic performance of noninvasive testing in symptomatic patients, recognizing that symptoms after surgery are often non-specific. Finally, although we could evaluate for different threshold effects of CRP and fecal calprotectin concentrations on diagnostic accuracy, this was more challenging to apply for radiographic studies where definitions of CTE/MRE/IUS recurrence varied between authors.

Conclusion

In summary, this systematic review and meta-analysis defines the diagnostic utility of CRP, fecal calprotectin, and bowel imaging for detecting CD recurrence after resection with primary ileocolonic anastomosis. Future work should focus on optimizing radiographic definitions of recurrence in the post-surgical bowel and evaluate whether the combination of different diagnostic modalities provides sufficient confidence for clinical decision-making in lieu of endoscopic assessment.

Supplementary Material

Note: To access the supplementary material accompanying this article, visit the online version of *Clinical Gastroenterology and Hepatology* at www.cghjournal.org, and at https://doi.org/10.1016/j.cgh.2025.03.030.

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Acknowledgments
The authors thank Caitlin McClurg (medical librarian) for her assistance in designing and executing the systematic review search.

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Priyanka Gill (Data curation: Supporting; Writing – review & editing: Supporting)
Levi R. Stein (Data curation: Supporting; Writing – review & editing: Supporting)
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Rupert W. Leong (Writing – review & editing: Supporting)
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John K. Marshall (Writing – review & editing: Supporting)
Siddharth Singh (Writing – review & editing: Supporting)
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Kerri L. Novak (Writing – review & editing: Supporting)
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Conflicts of interest
These authors disclose the following: Rupert W. Leong has served as an advisory board member of AbbVie, Aspen, BMS, Celgene, Celltrion, Chiesi, Ferring, Glutagen, Hospira, Janssen, Lilly, MSD, Novartis, Pfizer, Prometheus Biosciences, and Takeda; and is also a research grant recipient from Celltrion, Shire, Janssen, Takeda, Gastroenterological Society of Australia, National Health and Medical Research Council, Gutsy Group, Pfizer, and Joanna Tiddy grants at the University of Sydney. Peter De Cruz has served as a consultant, advisory board member or speaker for AbbVie, Baxter, Ferring, Janssen, Takeda, Celltrion and Emerge Health; is supported by a National Health and Medical Research Council Emerging Leadership Level 2 Fellowship; and has received research support from AbbVie, Ferring, Shire, Janssen, Takeda, and Celltrion. Gilaad G. Kaplan has received honoraria for speaking or consultancy from AbbVie, Amgen, Janssen, Pfizer, and Takeda; has received grants for research from Ferring and for educational activities from AbbVie, Bristol Myers Squibb, Ferring, Fresenius-Kabi, Janssen, Pfizer, and Takeda; and shares ownership of a patent: Treatment of Inflammatory Disorders, Autoimmune Disease, and PBC. UTI Limited Partnership, assignee. Patent WO2019046959A1. PCT/CA2018/051098. 7 Sept. 2018. Cyntha H. Seow reports advisory boards for Janssen, Abbvie, Takeda, Lilly, Ferring, Shire, Pfizer, Sandoz, Pharmascience, Fresenius Kabi, Amgen, Bristol Myers Squibb, and Celltrion; and speaker for Janssen, Abbvie, Takeda, Lilly Ferring, Shire, Pfizer, Pharmascience, and Fresenius Kabi. Cathy Lu has received speaker fees from Abbvie, Janssen, and Fresenius Kabi; and has received advisory board fees from AbbVie, Celltrion, Janssen, Lilly, Pfizer, Takeda, Fresenius Kabi, Pendopharm, and Ferring. Frank Hoentjen has served on advisory boards or as speaker for Abbvie, Janssen, MSD, Takeda, Pfizer, Celltrion, Teva, Sandoz, Amgen, and Pendopharm; and has received independent research funding from Janssen, Abbvie, Pfizer, and Takeda. John K. Marshall has received honoraria for speaking and/or consulting from AbbVie, Alimentiv, Amgen, Avir Pharma, Bausch Health, Bristol Myers Squibb, Celltrion, Ferring, Fresenius Kabi, Janssen, Lilly, Organon, Pfizer, Pharmascience, Regeneron, Roche, Sandoz, Sanofi, Takeda, and Viatris. Siddharth Singh has received a research grant from Pfizer. Remo Panaccione reports consultant for Abbott, AbbVie, Abbivax, Alimentiv (formerly Robarts), Amgen, AnaptysBio, Arena Pharmaceuticals, AstraZeneca, Biogen, Boehringer Ingelheim, Bristol-Myers Squibb, Celgene, Celltrion, Cosmos Pharmaceuticals, Eisai, Elan, Eli Lilly, Ferring, Galapagos, Fresenius Kabi, Genentech, Gilead Sciences, Glaxo-Smith Kline, JAMP Bio, Janssen, Merck, Mylan, Novartis, Oppilan Pharma, Organon, Pandion Pharma, Pendopharm, Pfizer, Progenity, Prometheus Biosciences, Protagonist Therapeutics, Roche, Sandoz, Satisfai Health, Shire, Sublimity Therapeutics, Spyre Therapeutics, Takeda Pharmaceuticals, Theravance Biopharma, Trellus, Union Biopharma, Viatris, Ventyx, and UCB; speaker's fees from AbbVie, Amgen, Arena Pharmaceuticals, Bristol-Myers Squibb, Celgene, Eli Lilly, Ferring, Fresenius Kabi, Gilead Sciences, Janssen, Merck, Organon, Pfizer, Roche, Sandoz, Shire, and Takeda Pharmaceuticals; and advisory boards for AbbVie, Alimentiv (formerly Robarts), Amgen, Arena Pharmaceuticals, AstraZeneca, Biogen, Boehringer Ingelheim, Bristol-Myers Squibb, Celgene, Eli Lilly, Ferring, Fresenius Kabi, Genentech, Gilead Sciences, Glaxo-Smith Kline, JAMP Bio, Janssen, Merck, Mylan, Novartis, Oppilan Pharma, Organon, Pandion Pharma, Pfizer, Progenity, Protagonist Therapeutics, Roche, Sandoz, Shire, Sublimity Therapeutics, Takeda Pharmaceuticals, and Ventyx. Kerri L. Novak has received educational grants from Janssen, Pfizer, Takeda, and Abbvie; and honoraria from Janssen Pfizer Takeda, Abbvie, Ferring, Lilly, Bristol Myers Squibb, Amgen, and Satisfai Health. Christopher Ma has received consulting fees from AbbVie, Alimentiv, Amgen, AVIR Pharma Inc, Bristol Myers Squibb, Celltrion, Eli Lilly, Ferring, Forte Biosciences, Fresenius Kabi, Janssen, McKesson, Mylan, Pendopharm, Pfizer, Prometheus Biosciences Inc, Roche, Sanofi, Takeda, and Tillotts Pharma; speaker's fees from AbbVie, Amgen, AVIR Pharma Inc, Alimentiv, Bristol Myers Squibb, Eli Lilly, Ferring, Fresenius Kabi, Janssen, Organon, Pendopharm, Pfizer, Sanofi, Takeda, and Tillotts Pharma; royalties from Springer Publishing; and research support from AbbVie, Eli Lilly, Ferring, Pfizer. The remaining authors disclose no conflicts.

Supplemental Appendix 1. Search Strategy

Embase (OVID)

1 exp Crohn disease/

2 crohn*.tw,kf.

3 1 or 2

4 (surg* or pre-surg* or presurg* or post-surg* or postsurg* or perisurg* or peri-surg* or post-operat* or postoperat* or peri-operat* or perioperat* or pre-operat* or preoperat*).tw,kf.

5 exp colectomy/

6 exp abdominal surgery/

7 Digestive System Surgical Procedures.tw,kf.

8 exp proctocolectomy/

9 exp anastomosis/

10 (resect* or colectom* or strictureplast* or operat* or proctocolectomy or ileal pouch-anal anastomosis or IPAA or anastomosis or balloon).tw,kf.

11 5 or 6 or 7 or 8 or 9 or 10

12 exp C reactive protein/

13 exp lactoferrin/

14 exp biological marker/

15 (c-reactive protein or lactoferrin or calprotectin or marker* or biomarker* or predict* or clinical or labrator* or serol* or serum or fecal* or feces or blood or cytokine*).tw,kf.

16 exp nuclear magnetic resonance imaging/

17 exp x-ray computed tomography/

18 exp echography/

19 (magnetic resonance imag* or enterograph* or colongraph* or mri or imag* or computed tomograph* or ultraso*).tw,kf.

20 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19

21 exp recurrent disease/

22 recur*.tw,kf.

23 relaps*.tw,kf.

24 21 or 22 or 23

25 exp cohort analysis/

26 exp "randomized controlled trial (topic)"/

27 exp case control study/

28 (cohort* or trial or RCT).mp. or control.tw,kf. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword heading word, floating subheading word, candidate term word]

29 25 or 26 or 27 or 28

30 3 and 11 and 20 and 24 and 29

31 limit 30 to human

MEDLINE (OVID)

1 exp Crohn Disease/

2 crohn*.tw,kf.

3 1 or 2

4 (surg* or pre-surg* or presurg* or post-surg* or postsurg* or perisurg* or peri-surg* or post-operat* or postoperat* or peri-operat* or perioperat* or pre-operat* or preoperat*).tw,kf.

5 exp Colectomy/

6 exp Digestive System Surgical Procedures/

7 exp Proctocolectomy, Restorative/

8 exp Anastomosis, Surgical/

9 (resect* or colectom* or strictureplast* or operat* or proctocolectomy or ileal pouch-anal anastomosis or IPAA or anastomosis or balloon).tw,kf.

10 4 or 5 or 6 or 7 or 8 or 9

11 exp C-Reactive Protein/

12 exp Lactoferrin/

13 exp Biomarkers/

14 (c-reactive protein or lactoferrin or calprotectin or marker* or biomarker* or predict* or clinical or labrator* or serol* or serum or fecal* or feces or blood or cytokine*).tw,kf.

15 exp Magnetic Resonance Imaging/

16 exp Tomography, X-Ray Computed/

17 exp Ultrasonography/

18 (magnetic resonance imag* or enterograph* or colongraph* or mri or imag* or computed tomograph* or ultraso*).tw,kf.

19 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18

20 exp Recurrence/

21 recur*.tw,kf.

22 relaps*.tw,kf.

23 20 or 21 or 22

24 3 and 10 and 19 and 23

25 exp Cohort Studies/

26 exp Randomized Controlled Trials as Topic/

27 exp Case-Control Studies/

28 (cohort* or randomized control* trial or RCT).mp. or case control.tw,kf. [mp=title, book title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms]

29 25 or 26 or 27 or 28

30 24 and 29

31 limit 30 to humans

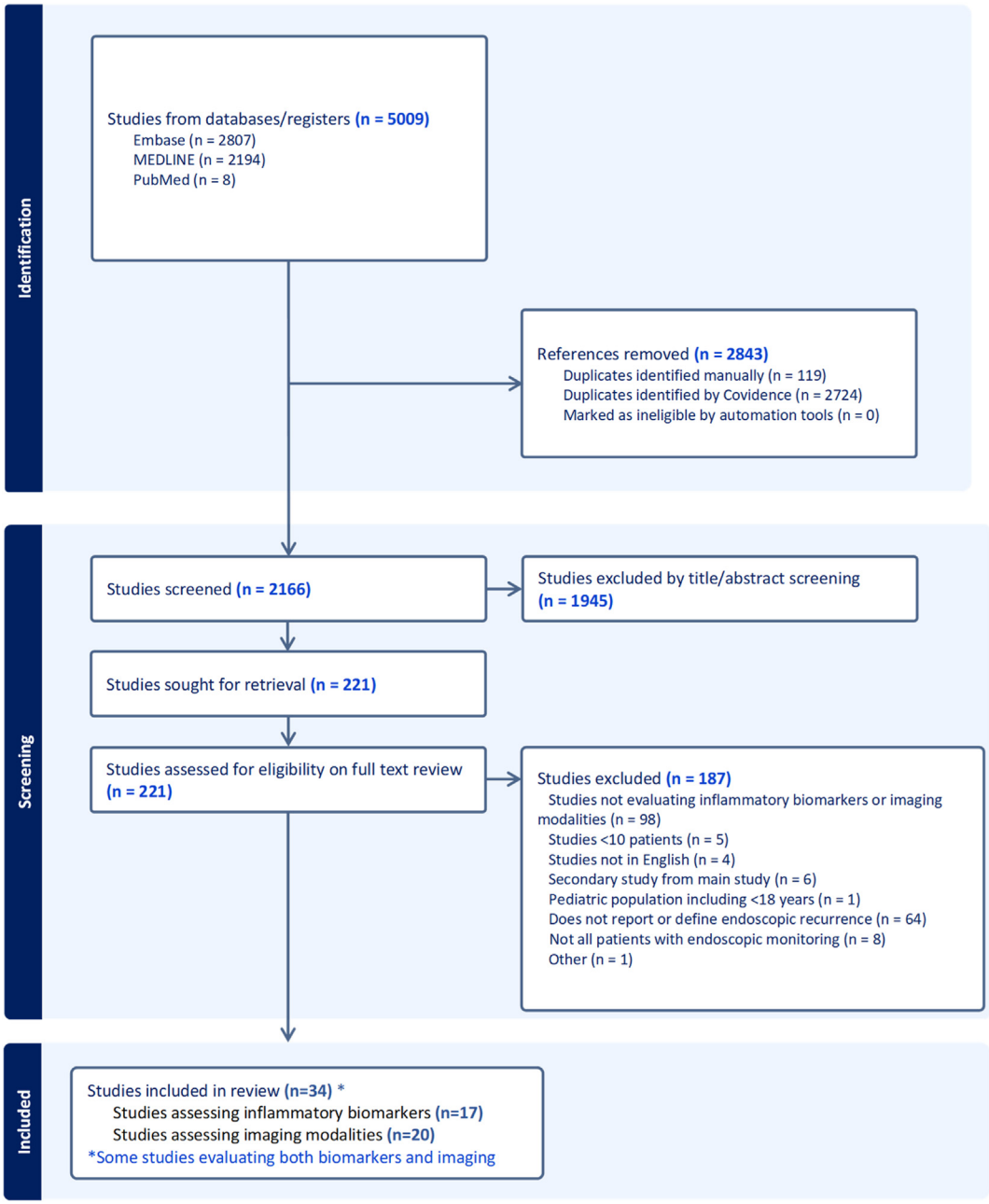
Supplemental Appendix 2. Risk of Bias Assessment (QUADAS-2)

Biomarker studies First author (year)	Item 1: risk of bias in patient selection	Item 2: representative patient spectrum?	Item 3: risk of bias in conduct or interpretation of index test	Item 4: applicability of index test to review question	Item 5: risk of bias in conduct or interpretation of reference standard	Item 6: risk of bias from applicability of reference standard	Item 7: risk of bias from timing/ patient flow
Lopes (2016)	Low	Low	Low	Low	Low	Low	Low
Bachiller (2016)	Low	Low	Low	Low	Low	Low	Low
Garcia-Planella (2016)	Low	Low	Low	Low	Low	Low	Low
Baillet (2018)	Low	Low	Low	Low	Low	Low	Low
Boube (2020)	Low	Low	Low	Low	Low	Low	Low
Yamamoto (2016)	Low	Low	Low	Low	Low	Low	Low
Wright (2015)	Low	Low	Low	Low	High	High	Low
Veyre (2021)	Low	Low	Low	Low	Low	Low	Low
Boschetti (2015a)	Low	Low	Low	Low	Low	Low	Low
Verdejo (2018)	Low	Low	Low	Low	Low	Low	Low
Daoud (2022)	Low	Low	Low	Low	Low	Low	Low
Orlando (2006)	Low	Low	Unclear	Unclear	Low	Low	Unclear
Regueiro (2011)	Unclear	Low	Low	Low	Low	Low	Unclear
Lasson (2014)	Low	Low	Low	Low	Low	Low	Low
Lobaton (2013)	Low	Low	Low	Low	Low	Low	Low
Furfaro (2023)	Low	Low	Low	Low	Low	Low	Low
Boschetti (2015b)	Low	Low	Low	Low	Low	Low	Low

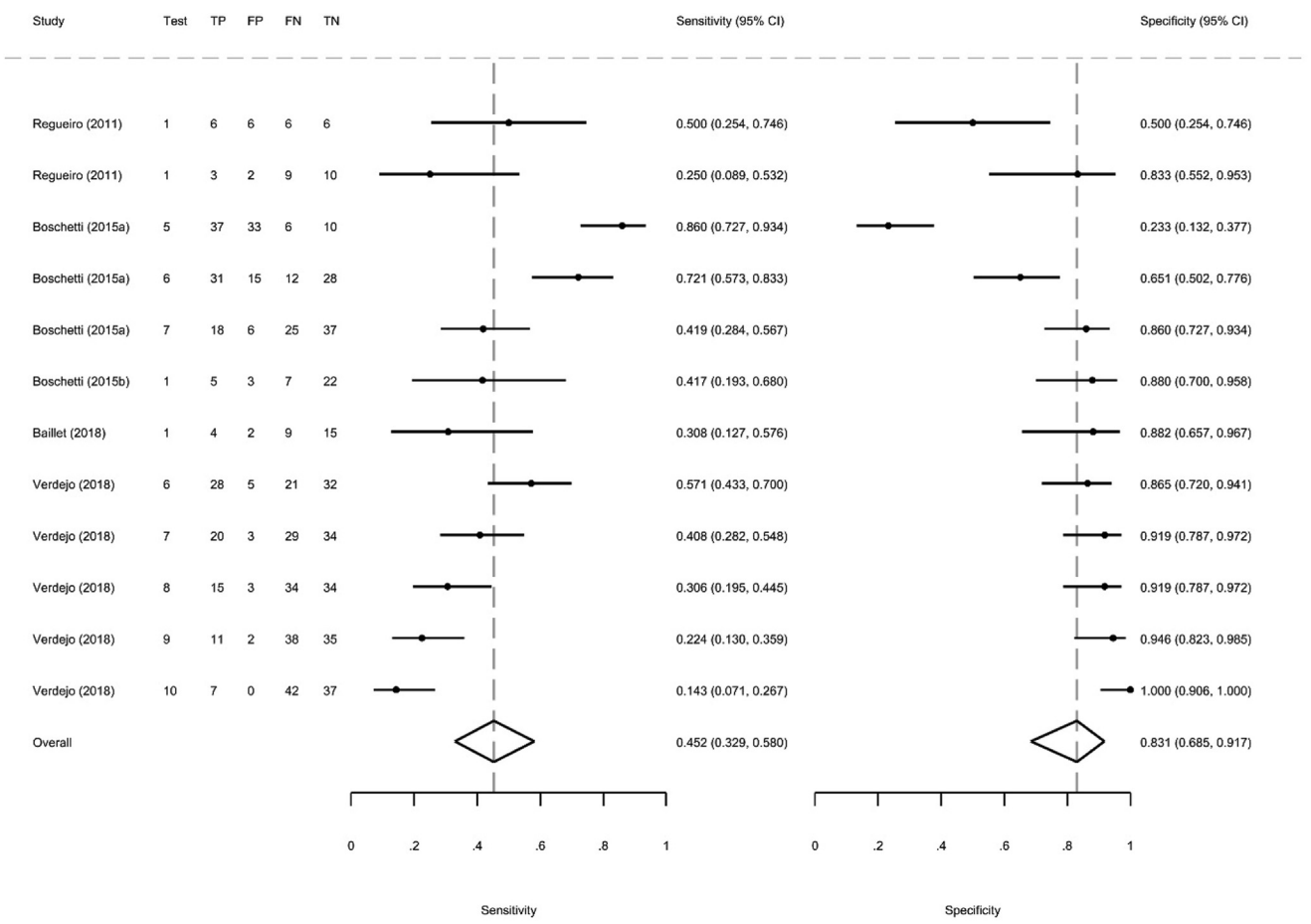
Imaging studies First author (year)	Item 1: risk of bias in patient selection	Item 2: representative patient spectrum?	Item 3: risk of bias in conduct or interpretation of index test	Item 4: applicability of index test to review question	Item 5: risk of bias in conduct or interpretation of reference standard	Item 6: risk of bias from applicability of reference standard	Item 7: risk of bias from timing/ patient flow
Bachour (2022)	Low	Low	Low	Low	Low	Low	Low
Castiglione (2008)	Low	Low	Low	Low	Low	Low	Low
Furfaro (2023)	Low	Low	Low	Low	Low	Low	Low
Pozzessere (2022)	Low	Low	Low	Low	Low	Low	Low
Martinez (2018)	Low	High	Low	Unclear	Low	Low	Low
Onali (2010)	Low	Low	Low	Low	Low	Low	Low
Mao (2013)	High	Low	Low	Low	High	High	Low
Gallego Ojea (2010)	Unclear	Unclear	Unclear	Unclear	Low	Low	Low
Carmona (2021)	Low	Low	Low	Low	Low	Low	Low
Paredes (2013)	Low	Low	Low	Low	Low	Low	Low

Continued

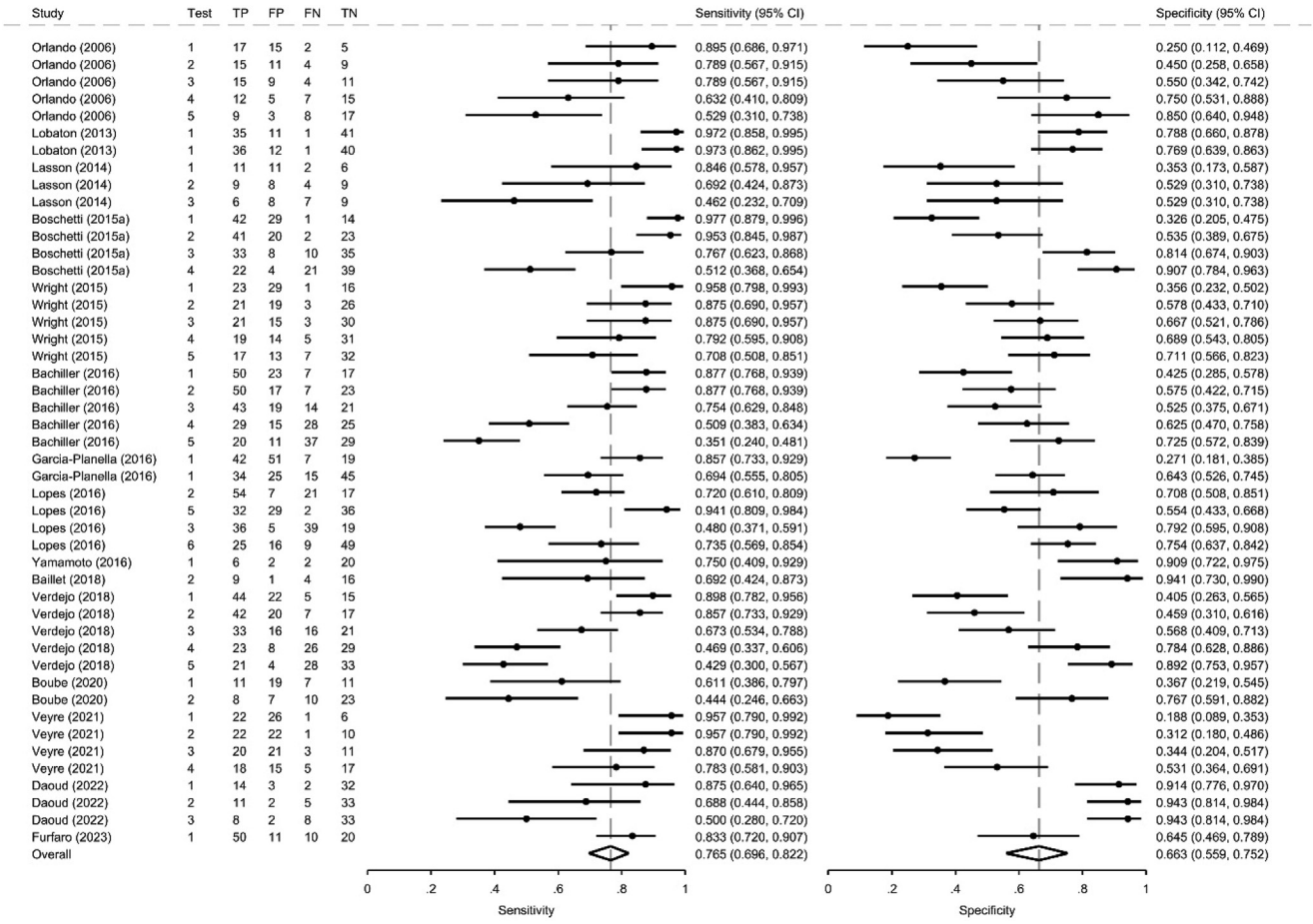
Imaging studies First author (year)	Item 1: risk of bias in patient selection	Item 2: representative patient spectrum?	Item 3: risk of bias in conduct or interpretation of index test	Item 4: applicability of index test to review question	Item 5: risk of bias in conduct or interpretation of reference standard	Item 6: risk of bias from applicability of reference standard	Item 7: risk of bias from timing/ patient flow
Paredes (2010)	Low	Low	Low	Low	Low	Low	Low
Andreoli (1998)	Unclear	Unclear	Low	Low	High	High	Low
Calabrese (2009)	Low	Unclear	Low	Low	Low	Low	High
Rispo (2006)	Unclear	Low	High	Low	Low	Low	Unclear
Aguas (2012)	Unclear	Unclear	Low	Low	Low	Low	Low
Baillet (2018)	Low	Low	Low	Low	Low	Low	Low
Orlando (2006)	Low	Low	Low	Low	Low	Low	Low
Onali (2016)	Low	Low	Low	Low	Low	Low	Low
Biancone (2007)	Low	Low	Low	Low	Low	Low	Low



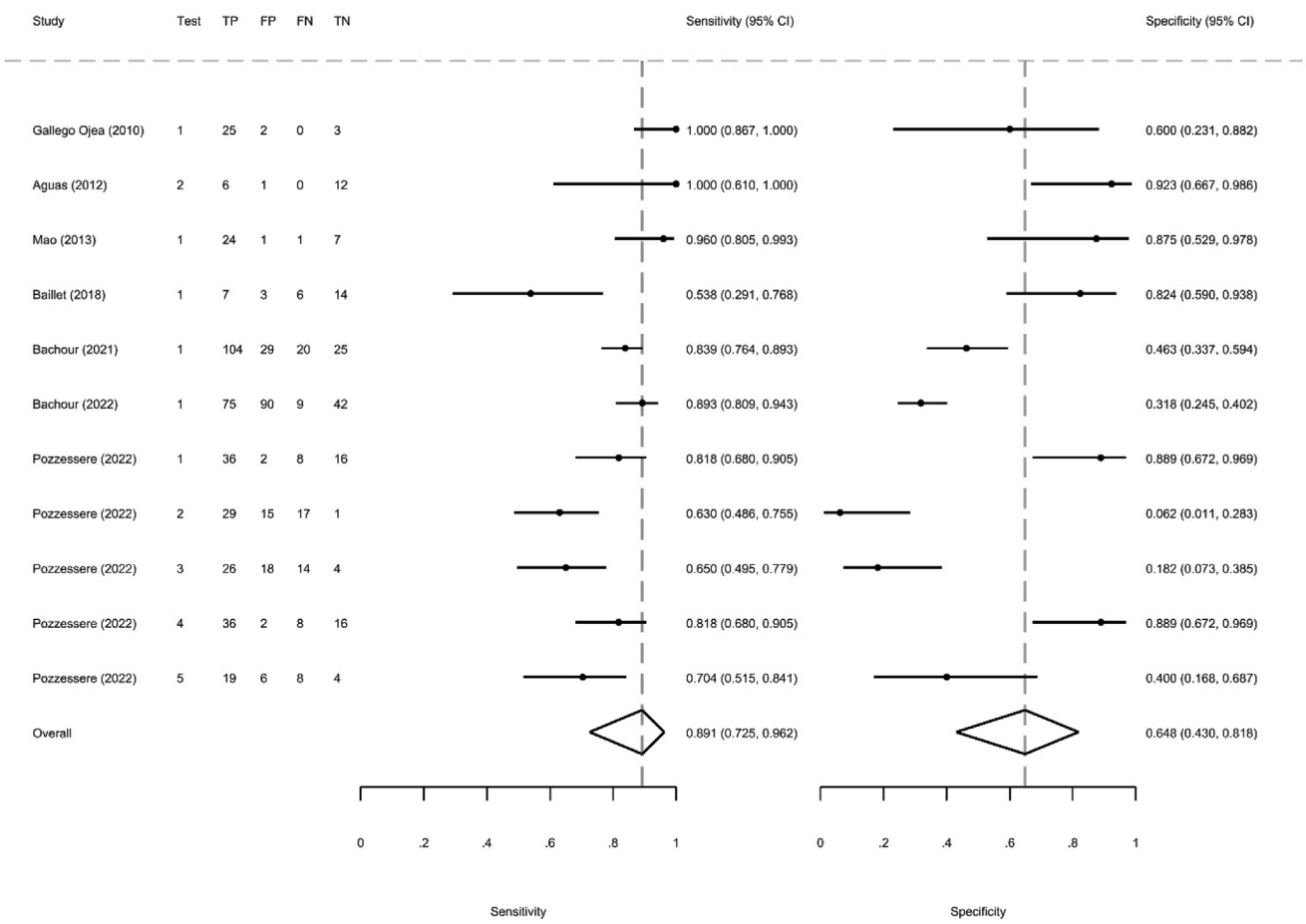
Supplementary Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) diagram.



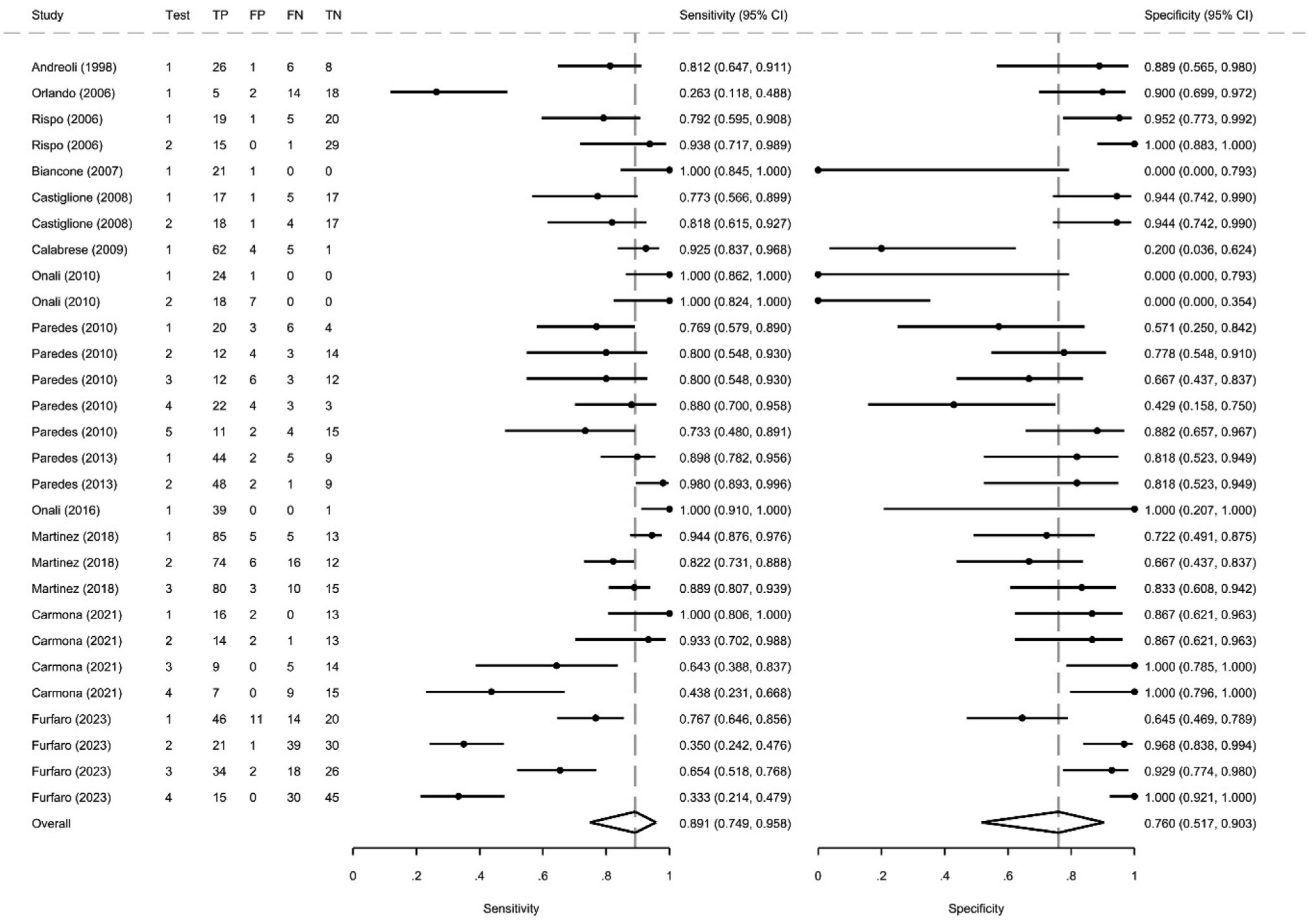
Supplementary Figure 2. Pooled sensitivity and specificity of CRP for detecting endoscopic recurrence of Crohn's disease.



Supplementary Figure 3. Pooled sensitivity and specificity of fecal calprotectin for detecting endoscopic recurrence of Crohn's disease. FN, false negative; FP, false positive; TN, true negative; TP, true positive.



Supplementary Figure 4. Pooled sensitivity and specificity of CTE or MRE for detecting endoscopic recurrence of Crohn's disease. FN, false negative; FP, false positive; TN, true negative; TP, true positive.



Supplementary Figure 5. Pooled sensitivity and specificity of intestinal ultrasound for detecting endoscopic recurrence of Crohn's disease. FN, false negative; FP, false positive; TN, true negative; TP, true positive.